ELLIS Winter School 2026:
AI for Earth System, Hazards & Climate Extremes

16 - 20 March, 2026
National Technical University of Athens

Patission Complex

PROGRAMME

MONDAY 16/3

8:30 - 9:00
Registration, Welcome, Poster Set-up
Venue 1

9:00 - 9:45
Opening Speeches

- Prof. Konstantinos Karantzalos, General Secretary, Hellenic Ministry of Digital Governance & NTUA
- Prof. Matthaios Papavasiliou, Dean, School of Architecture, NTUA
- Prof. Vassilia Karathanassi, NTUA
- Asst. Prof. Ioannis Papoutsis, NTUA
- Dr. Nuno Carvalhais, Group Leader at Max Planck Institute for Biogeochemistry
- Dr. Conrad Philipp, Unit Coordinator, ELLIS Unit Jena

Venue 1

9:45 - 10:30
Keynote

"Early Warning of Complex Climate Risk - AI challenges and opportunities"
Prof. Markus Reichstein
Max Planck Institute for Biogeochemistry
Venue 1

10:30 - 11:00
Coffee Break
Venue 1

11:00 - 11:45
Keynote

"Building the Planetary Data Layer: AI and the Rise of the Observable Earth"
Martin Langer
CEO, OroraTech
Venue 1

11:45 - 12:30
Group Work Intro
Venue 1

12:30 - 14:00
Lunch & Poster Session
Venue 1

TUESDAY 17/3

8:30
Departure from Athens City Centre (Metro Omonoia)

9:30 - 9:45
Arrival to Lavrion

9:45 - 10:30
Beach Walk

10:30 - 11:00
Coffee Break

11:00 - 11:45
Keynote

"AI for Earth at Scale: From Scientific Breakthroughs to Societal Impact"
Sabrina Ricci
AI Ecosystem Coordinator, ESA phi-lab

11:45 - 12:30
Keynote

"Foundation models and their adaptation to earth observation"
Assistant Prof. Maria Vakalopoulou
Universite Paris-Saclay

12:30 - 12:45
Presentation

"Building the future on the ruins of the past"
Assimakis Chadoumellis
LTCP Site Manager

12:45 - 14: 15
Lunch & Venue Tour

14:15 - 15:00
Keynote

"Predicting and projecting the global carbon cycle and ocean compound extremes"
Dr. Hongmei Li
Senior Research Scientist, Hamburg & Helmholtz Centre Hereon

WEDNESDAY 18/3

8:30 - 9:00
Attendance & Poster Set-up
Venue 1

9:00 - 9:45
Keynote

"Beyond XAI: Causal Reasoning for Trustworthy Geoscience AI"
Prof. Gustau Camp-Valls
University of Valencia
Venue 1

9:45 - 10:30
Keynote

"Responsible AI in EO: Aligning Data, Models, and Principles"
Prof. Pedram Ghamisi
Helmholtz-Zentrum Dresden-Rossendorf
Venue 1

10:30 - 11:00
Coffee Break
Venue 1

11:00 - 12:30
Zero2Hero Sessions

"Datacubes"
Dr. Mohanad Albughdadi
ECMWF
Venue 1

"Hybrid modeling and parameterization ocean models"
Dr. Said Ouala
Institut Mines-Télécom Atlantique
Venue 2

12:30 - 14:00
Lunch & Poster Session
Venue 1

THURSDAY 19/3

8:30 - 9:00
Attendance & Poster Set-up
Venue 1

9:00 - 9:45
Keynote

"Learning from Variability: AI insights into Extreme events"
Ramon Fuentes
Researcher, SMHI
Venue 1

9:45 - 10:30
Keynote

"The contribution of satellite data and artificial intelligence for earthquake response with damage mapping at building level"
Mounia El Baz
EO Digital Innovation Engineer, ESA phi-Lab
Venue 1

10:30 - 11:00
Coffee Break
Venue 1

11:00 - 12:30
Zero2Hero Sessions

"Practical exercise with Causme platform"
Dr. Homer Durand
University of Valencia
Venue 1

"Machine learning for earth system science from the beginning or on a special aspect"
Dr. Christian Reimers
Max Planck Institute for Biogeochemistry
Venue 2

12:30 - 14:00
Lunch & Poster Session
Venue 1

FRIDAY 20/3

8:30 - 9:00
Attendance & Poster Set-up
Venue 1

9:00 - 9:45
Keynote
"Uncertainty Quantification in Geospatial Machine Learning Applications"
Alexander Brenning
Head of Chair "GIScience Group", University of Jena
Venue 1

9:45 - 10:30
Keynote

"Ethical use of Artificial Intelligence in Earth Observation: Issues and Insights from the EU AI Act"
Prof. Mrinalini Kochupillai
Hanken School of Economics
Venue 1

10:30 - 12:30
Coffee Break
Venue 1

11:00 - 11:45
Keynote

"Extremes in a changing climate — what we know and where we struggle"
Dr. Dominik Schumacher
ETH
Venue 1

11:45 - 12:00
Presentation

Topic TBA
Chrysa Oikonomou, Evangelos Kalamaras
ICEYE

All times are in local Greece time (EET)

Co-funded by the European Union